JianghuSect 2 (JS2-v2.6.0) — Deep-Dive Feature & Logic Audit

System #01 (Part 1 of 2): Cultivation Paths & Subpaths — Genesis, Path Library, Selection Math, Alias Integrity & Retraining Lifecycle

- **Target Codebase**: Js2 new (SIM.PATCH_VERSION = '2.6.0', SIM.SAVE_SCHEMA_VERSION = '2.6.0-living-chronicle-and-reporting', 43 Major Paths / 286 Subpaths)
- **Target Readers**: Autonomous Coding Models (**GPT** & **Gemini** Systems Architects / Developers)
- **Audit Scope (Part 1 of 2)**:
    1. Core Emergent Simulation Philosophy & Non-Negotiable Engine Invariants
    2. [PATH-GEN-01] Sect Founding Monoculture Lock, Prodigy Railroading & The "Mute Grandmaster" Paradox (sim.js, cultivation.js, sect.js)
    3. [PATH-LIB-02] Split-Brain Knowledge Gating (SECT.PATH_LIBRARY vs knowledge.js vs retraining.js) & Dead No-Op Guards (ai.js)
    4. [PATH-SEL-03] Corrupted Trait/Constitution Inputs into Path Selection & Elemental Access (data_core.js × human_core.js × cultivation.js)
    5. [PATH-ALIAS-04] Subpath Alias Collisions & Silent Identity Erasure (cultivation.js × martial.js × visitors_clues.js)
    6. [PATH-RET-05] Broken Retraining Track Initialization, Intra-Path Lifespan Death-Trap, Residual State Pollution & Flat Subpath Tiering (retraining.js, knowledge.js, fishing.js)
- **Companion Document (Part 2 of 2 — Next Pass)**: Asymmetric maxStage (4/5/8/9) Progression Math, effectiveTechStage Track Exploit, 4-Wave Daily AI & Station Routing Inequality (80 Unreachable Peak Stations), Era-5Dormant State Bug (mantraState/defensiveArtState/supportState/forbiddenArtState), Disembodied Monthly God-Scripts (monthlyAutonomousTick), and HIGHER_REALMS_PANTHEON Stat Inversion.

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0. Core Game Design Paradigm & Non-Negotiable Technical Invariants (Read First)

Before modifying any line of code in **JianghuSect 2**, every AI developer (GPT or Gemini) must internalize the game's foundational identity. Many existing logical bugs exist because previous patches implemented **RPG-style player-triggered mechanics** or **top-down scripted timers** instead of **bottom-up emergent NPC agency**.

0.1 The Four Pillars of Emergent Simulation Philosophy

1. **100% Autonomous Agent-Driven World (`NPC Autonomy First`)**:
   - Every action, cultivation choice, path awakening, subpath specialization, retraining pivot, vow, and world-system mechanic must originate from an individual NPC evaluating their own internal state (`npc.h` physiology, intellect, personality, values, affinities, spiritual roots, memories, relationships, injuries, lifespan pressure, and cognition goals).
2. **Zero Predetermined Locks (`Anti-Railroading`)**:
   - Never hard-lock a sect, world, or NPC into a rigid template. A sect founded as a Sword Sect can organically evolve into a Poison-Gu & Mercantile Consortium if its surviving elders, captured manuals, and talented recruits pull it in that direction.
3. **Player = God-Observer & Sandbox Editor Only (`Zero Direct Puppeteering`)**:
   - The player **cannot** command NPCs to perform actions, switch stances, unseal taboo arts, or choose paths via RPG-style control buttons. The player may only observe the living simulation or use sandbox editor tools (`SECT.EDITOR`) to edit stats/attributes or spawn items into inventories. Any feature that relies on a player clicking a UI button to activate an NPC's state is **dead code** in real gameplay.
4. **Physical & Causal Grounding (`No Disembodied God-Hands`)**:
   - Events must happen through physical stations, travel, interpersonal mentorship, or genuine utility-AI actions (`AI.ACTS` + `COG`), never via a monthly global loop grabbing `W.npcs[0]` regardless of whether they are asleep, imprisoned, away on an expedition, or ideologically opposed to the act.

0.2 Hard Technical Invariants

- **Locked Human Core**: `GAME_SOURCE/js/human_core.js` is strictly immutable (`HC.FINGERPRINT === '155:2f9c9dd7'`, SHA-256 `a0d393284fb4b136848e5b80e12f3edd2c4f9437c419c793c2a7e75d881c0542`). Never edit a single byte of `human_core.js`. Intercept or normalize data in downstream wrappers (`cultivation.js`, `martial.js`, `cognition.js`, `ai.js`).
- **Script Load Order**: Preserve the exact 40 `<script>` tags across `GAME_SOURCE/index.html` and `GAME_SOURCE/test/load.js`.

- **Closed-Loop Conservation (`Debit === Credit`)**: Spirit stones, manuals, crystals, and materials must never materialize from thin air or vanish into the void without explicit world-pool accounting.
- **Simulation Purity**: UI rendering, chronicle formatting, and read-only inspection (`STATS.explain`, `COG.evaluate*`) must consume **zero** `W.rng` calls and cause **zero** state mutations (`SIM.serialize(W)` byte-identical before and after).
- **Performance Ceiling**: `SIM.tick(W)` with 80 active NPCs must remain `<= 3.5 ms/tick`.
- **Anti-Stubbing Test Mandate**: Every fix must be verified by an end-to-end multi-tick behavioral test proving that an autonomous NPC discovers, decides, executes, and benefits from the mechanic in a live `SIM.tick` world—never merely checking `typeof fn === 'function'`.

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1. `[PATH-GEN-01]` Sect Founding Monoculture Lock, Prodigy Railroading & The "Mute Grandmaster" Paradox

1.1 Exact Code Anchors

- `js/sim.js`: `SIM.buildNpc` / `SIM.awaken` (lines 95–140), `SIM.masterOpts` / `SIM.candidates` (lines 147–174), `SIM.create` (lines 195–230), `SIM.addNpc` (lines 296–334)
- `js/cultivation.js`: `CULT.MORTAL_INITIATION` (`ensureMortalState`, `stepStudyDao`, `stepGatherQi`, `triggerMeridianBreakthrough`)
- `js/sect.js`: `SECT.PATH_LIBRARY` (`unlockedPaths`, `canDiscipleStudyPath`, `donatePathManual`, `bestowPathCrystal`)

1.2 Root-Cause Code & Logic Defects

Defect 1A — Blind Path Forcing on Founders & Master Candidates (`sim.js:97–115`)

In `SIM.awaken`:

```javascript
if (archDef && o.archetypeKey && W._founding !== false) {
  const primaryPath = archDef.path || 'XIAN';
  const samePathSubs = (archDef.subs || []).filter(...);
  const subIdx = (o.founderIndex != null ? o.founderIndex : 0) % validPool.length;
  const chosenSub = validPool[subIdx] || validPool[0];
  const memberPath = primaryPath;
  npc.cult = { path: memberPath, sub: chosenSub, stage: 1, ... };
```

```
}
```

Meanwhile, `SIM.buildNpc` generates the founder's `npc.h` (`HC.generate`) using unbiased random latent factors and `ELEM.rollBirthRoots` **without aligning or filtering against `archDef.path`**.

- **Concrete Failure**: If the player starts a `HEART_SOUL` (`LIGHT_YANG` / `ASTRAL`) or `SONIC_HARMONY` (`WIND`) sect, `HC.generate` frequently rolls a founder or founding disciple with a **Single Defective/Mortal Water or Earth Root**, `aggression: 88`, `comprehension: 22`, or `BODY_MARTIAL_ZEALOT` personality archetype. `SIM.awaken` ignores `CULT.pathScores(npc)` and forces them onto `memberPath` and `validPool[founderIndex % length]`.
- **Immediate Impact**: Under `ELEM.resolveCompatibility`, that founder suffers permanent `CLASH` or `POOR` compatibility (`0.60x` cultivation rate, `0.65x` skill damage, `1.30x` Qi cost, `+18%` breakthrough backlash risk) in the very path they supposedly founded!

Defect 1B — Total Monoculture Railroading of Mortal Prodigies (`CULT.MORTAL_INITIATION`)

At world creation, `W.sect.pathLibrary.unlockedPaths` contains **exactly 1 path** (`[arch.path]`). Every mortal recruit (`rank === 0`, `fromApplicant: true`) initializes `mortalCult.targetPath` exclusively from `W.sect.pathLibrary.unlockedPaths`.

- **Why It Violates Context & Emergent Agency**:
  1. Suppose a once-in-a-millennium genius (`HEAVENLY` Lightning Root + `CONST_IRON_VAJRA` + `TR_IN_CRIPPLED_ROOT` whose lore explicitly says "เหมาะกับการเปลี่ยนไปฝึกวิถีกายา") joins a `VENOM_GU` or `MERCANTILE_DAO` sect.
  2. The mortal has **zero autonomous ability** to recognize that `VENOM_GU` clashes violently with their body/root, zero ability to delay Dantian awakening while searching for a compatible primer, zero ability to self-awaken an innate physical/body path (`BODY` / `ARENA_GLORY`), and zero ability to request or purchase a `CRYSTAL_PATH_*` sitting in the sect warehouse!
  3. Instead, `A.study_dao` and `A.gather_qi` robotically force them into `VENOM_GU` after ~60 days, permanently ruining a 1,000-year prodigy and forcing them to later sacrifice half their cultivation stages if they ever retrain.

Defect 1C — Asymmetric Awakening Bypass in `SIM.addNpc` (`sim.js:305–315`) vs The "Mute Grandmaster" Paradox

Look at `SIM.addNpc`:

```javascript
const awaken = o.fromExped ? 0.65 + Math.max(0, effQuality) * 0.15 : 0.5;
const isMortalRecruit = o.stage == null && ((o.rank || 0) === 0 ||
o.fromApplicant) && !o.fromExped && !o.forceAwaken;
```

When an already-awakened outsider joins the sect (via expedition rescue `fromExped: true`, wandering figure recruitment `EXPED.recruitWanderingFigure`, prison defection `RECRUIT_DEFECTOR`, or `CRYSTAL_PATH_*`), they enter `W.npcs` practicing one of the other 42 Major Paths (e.g., Stage 6 `CHRONO_EPOCH`).

- **The "Mute Grandmaster" Paradox**:
  - That Stage 6 `CHRONO_EPOCH` elder can take personal disciples (`mentor`), become Peak Master, or even succeed as **Sect Master** (`W.sect.masterId`) if the original founder dies!
  - Yet because `W.sect.pathLibrary.unlockedPaths` is **only** mutated by consuming a physical `MANUAL_PATH_CHRONO_EPOCH` item, **a Stage 6–9 Grandmaster of `CHRONO_EPOCH` serving as Sect Master cannot author a basic Stage-1 introductory Dao primer (`MANUAL_PATH_CHRONO_EPOCH`) for their own sect's Scripture Pavilion, nor initiate their own personal mortal apprentice into `CHRONO_EPOCH`!**
  - Every mortal in the `CHRONO_EPOCH` Sect Master's sect continues awakening exclusively into the dead former master's path!

Defect 1D — Orphaned `CRYSTAL_PATH_*` Stock in Autonomous Play

All 43 `CRYSTAL_PATH_<PATH>` items exist in `ECO.itemById` and drop from Grand Dao Debates, Pagoda Apex first-clears, caravans, and expeditions into `W.sect.stock`.

- **Broken Loop**: Neither `BN.daily`, `CPE.welfareDaily`, `CPE.pickPurchase`, `COG.prepare`, nor any officer duty (`scripture` / `registry` / `master`) **ever** evaluates matching a `CRYSTAL_PATH_<PATH>` in `W.sect.stock` to a mortal disciple whose root/aptitude strongly matches `<PATH>`! Thus, `CRYSTAL_PATH_*` items pile up unused in `W.sect.stock` forever unless spawned/triggered via external test scripts or player intervention.

1.3 Multi-Order Chain Reactions (Now & Long-Horizon Year 10–50)

- **1st-Order**: Early-game sects become 100% path clones regardless of the diverse genetic/root population entering through `SECT.rollApplicantTier`.
- **2nd-Order**: Cross-Path Synergies (`COMBAT.CROSS_PATH_SYNERGIES`), Multi-Path Duet Skills (`COMBAT.DUET_CATALOG`), and 8-Peak Specialization (`SECT.PEAKS`) remain largely inert in natural runs unless random caravan RNG happens to sell a `MANUAL_PATH_*` before treasury reserve floors block purchases.

- **3rd-Order (Succession & Dynasty Collapse)**: When a recruited outsider elder inherits leadership after a raid wipes out the founding lineage, the sect suffers permanent ideological schizophrenia—its living leadership practices Path B, its library and mortals are frozen in Path A, and mentorship compatibility (`COG.completeMentorLesson` / `SECT.MENTORSHIP.transmitQi`) breaks down across generations.

1.4 Architectural Remediation Blueprint for AI Developers

1. **Candidate & Founder Bias Alignment (Not Post-Hoc Overwrite)**:
    - In `SIM.masterOpts` / `SIM.buildNpc` during founding (`o.isFounder`), pass mild root/affinity priors matching `archDef.path` (or pick the best-fitting candidate from a small deterministic sample of 3 rolls using `r`) rather than generating a Fire-hating Water root and brute-forcing them into a Fire path.
2. **Autonomous Elder Primer Authorship (`Oral Lineage -> Written Canon`)**:
    - Add an autonomous Scripture Pavilion / Elder activity (e.g., `codifyPathPrimer` or inside `scripture` / `master` / `elder` cognition goals): any loyal (`loyalty >= 60`), high-realm (`equivalentStage >= 5`, `SCRIPTURE >= 45`) member whose `cult.path` is **not** yet in `W.sect.pathLibrary.unlockedPaths` can spend time at `scripture` + real ink/cinnabar/spirit-stone cost (`Debit === Credit`) to transcribe `MANUAL_PATH_<theirPath>` into the Sect Path Library!
    - Likewise, a direct mentor (`n.mentor`) of `equivalentStage >= 4` should be able to guide their sworn mortal `disciple` into `mentor.cult.path` via direct oral transmission (`daoComprehension` gained from `pair teach` sessions), even before a public manual exists in the library.
3. **Mortal Path Holdout & Autonomous `CRYSTAL_PATH_*` Allocation**:
    - In `CULT.MORTAL_INITIATION`, evaluate `CULT.majorPathScores(mortal)` across all 43 paths vs the best unlocked path in `W.sect.pathLibrary.unlockedPaths`:
        - If `W.sect.stock['CRYSTAL_PATH_' + topNaturalPath] > 0` (or in `mortal.inv`), allow the `scripture`/`registry`/`master` officer (weighted by `meritocracy` vs `favoritism` in `COG`)—or the mortal via `CPE.pickPurchase`—to bestow/exchange that `CRYSTAL_PATH_*` autonomously!
        - If a mortal has a severe `CLASH` with every unlocked sect path AND high `willpower`/`pride` (or `TR_IN_CRIPPLED_ROOT` / `BODY_MARTIAL_ZEALOT`), allow them to either self-awaken into primal `BODY` via arduous physical training (`practice` / `lift` accumulating `bodyFoundation`) or delay `gather_qi` (`limit` scaled by `patience`) while prioritizing exploration/clue/debate support.

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2. [PATH-LIB-02] Split-Brain Knowledge Gating (SECT.PATH_LIBRARY vs knowledge.js vs retraining.js) & Dead No-Op Guards

2.1 Exact Code Anchors

- `js/ai.js`: `A.study.done` (line ~560)
- `js/knowledge.js`: `branches` (line 116), `M.eligible` (lines 140–160), `M.donateManual` (lines 193–221), `M.rollHierarchicalDiscovery` (lines 242–249)
- `js/retraining.js`: `M.retrainingTarget` (lines 6–12), `M.retrainingQuote` (lines 13–23), `M.retrain` (lines 24–41)

2.2 Root-Cause Code & Logic Defects

Defect 2A — Sham / Dead No-Op Call in `ai.js`

In `js/ai.js` inside `A.study.done`:

```javascript
if (n.cult && g.SECT && g.SECT.PATH_LIBRARY && typeof
g.SECT.PATH_LIBRARY.canDiscipleStudyPath === 'function') {
  g.SECT.PATH_LIBRARY.canDiscipleStudyPath(W, n, n.cult.path);
}
```

The return value is **completely discarded**. It performs zero validation, zero branching, and zero logging. It was clearly inserted only to pass a static substring search test.

Defect 2B — Complete Bypass of `SECT.PATH_LIBRARY` via `W.sect.subLibrary` and `retraining.js`

There are currently **two disconnected knowledge-library architectures** operating in parallel without mutual awareness:

1. **System 1 (`v2.0.0 SECT.PATH_LIBRARY`)**: Tracks unlocked Major Paths in `W.sect.pathLibrary.unlockedPaths` (unlocked by `MANUAL_PATH_<PATH>`). Claims in both documentation and `v200_jianghusect2_grand_dao_epoch_suite.js` (lines 72–74): *"Test subpath switching gate: switching to an unlearned path must fail — `SECT.PATH_LIBRARY.canDiscipleStudyPath(W, disc, lockedPath) === false`"*.

2. **System 2 (`v1.12.0` `knowledge.js` + `retraining.js`)**: Tracks unlocked Subpaths in `W.sect.subLibrary` (unlocked by `subManual` items) and one-use Subpath Crystals in `W.sect.crystals['S:<SUB>']`.
    - Look at `retraining.js:13–22` (`M.retrainingQuote`):

```javascript
const id = target.knowledge.id,
  ownedCrystal = (n.inv && n.inv.crystals && n.inv.crystals[key] || 0) > 0,
  ownedManual = (n.inv && n.inv.subManuals || []).includes(id),
  sectManual = (W.sect.subLibrary || []).includes(id) && SECT.canLearnTier(W, n)
>= target.knowledge.tier;
```

    - **Notice what is missing**: `M.retrainingQuote` **never calls `SECT.PATH_LIBRARY.canDiscipleStudyPath(W, n, target.knowledge.path)`** when `source === 'sectManual'` or `'personalManual'`!
    - Meanwhile, `M.rollHierarchicalDiscovery` (`knowledge.js:243`) and `FISHING.openContainer` (`fishing.js:597`) drop `subManual`s for all 286 subpaths across all 43 paths. As soon as any disciple donates a `subManual` (e.g., `SUNDIAL_EPOCH_CHRONICLER` of `CHRONO_EPOCH`) into `W.sect.subLibrary`, **every disciple in the sect can infinitely retrain into `CHRONO_EPOCH` via `sectManual` even though `CHRONO_EPOCH` is still LOCKED in `W.sect.pathLibrary.unlockedPaths`!**
    - Conversely, if `CHRONO_EPOCH` **is** unlocked via `MANUAL_PATH_CHRONO_EPOCH` (`W.sect.pathLibrary.unlockedPaths.includes('CHRONO_EPOCH')`), `W.sect.subLibrary` is **not** updated with starter subpaths of `CHRONO_EPOCH`, so existing disciples **still cannot retrain into `CHRONO_EPOCH`** via `retraining.js`! That means `MANUAL_PATH_*` and `subLibrary` contradict each other in **both directions**!

Defect 2C — Redundant Dual-Crystal Economy (`CRYSTAL_PATH_*` vs `S:<SUB>`)

- `CRYSTAL_PATH_<PATH>` lives in `W.sect.stock` / `n.inv.mats` (or consumable stock), priced in Spirit Stones, invisible to `CPE.shopItems`.
- `S:<SUB>` lives in `W.sect.crystals` / `n.inv.crystals`, priced in Contribution Points (`CP`), visible to `CPE.shopItems` and `M.daily`.
- Having two separate crystal items that both grant one-time cross-path/subpath access—stored in different inventory dictionaries, priced in different currencies, and governed by different eligibility functions—creates severe cognitive fragmentation for NPCs and future AI patches.

2.3 Architectural Remediation Blueprint for AI Developers

1. **Unify `SECT.PATH_LIBRARY` and `W.sect.subLibrary` Semantics Clearly**:
    - **Rule of Canon**:
        - `W.sect.pathLibrary.unlockedPaths` = **Foundational Major Path Canon** (allows mortals to awaken into `<PATH>` via `study_dao`, and allows existing cultivators to retrain into `<PATH>`'s **foundational Tier-1 subpath**—or unlocks `<PATH>` as a prerequisite for studying reusable `subManual`s in `W.sect.subLibrary`).
        - A reusable `subManual` in `W.sect.subLibrary` for a locked Major Path `<PATH>` should either:
            - **(Option A — Strict Hierarchy, Recommended)**: Require `<PATH>` to be unlocked in `W.sect.pathLibrary.unlockedPaths` before it can be taught as a reusable sect-wide curriculum (`sectManual`), **while** a single-use `S:<SUB>` crystal or `CRYSTAL_PATH_<PATH>` bypasses `unlockedPaths` for 1 individual disciple (matching the exact lore of one-use inheritance crystals!).
            - Furthermore, when `MANUAL_PATH_<PATH>` unlocks `<PATH>` in `SECT.PATH_LIBRARY`, automatically register `<PATH>`'s primary foundational subpath (`CULT.PATHS[path].subPaths[0].id`) into `W.sect.subLibrary` so existing cultivators can actually retrain into the newly unlocked Major Path!
2. **Bridge `CRYSTAL_PATH_*` into NPC Cognition & CP Shop**:
    - Expose `CRYSTAL_PATH_*` in `W.sect.stock` to `CPE.shopItems` / `CPE.pickPurchase` (for mortals with clashing roots or cultivators seeking cross-path retraining) and remove the dead no-op call in `ai.js:560`.

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3. [PATH-SEL-03] Corrupted Trait/Constitution Inputs into Path Selection & Special Elemental Access

3.1 Exact Code Anchors

- `data/data_core.js`: `curatedTraits` (`TR_IN_CRIPPLED_ROOT`, `TR_BL_POISON_IMMUNE`, conditional traits, `behaviorMods`), `constitutions` (`CONST_*`)
- `js/human_core.js`: `collectMods` (lines 109–118 — **IMMUTABLE, DO NOT EDIT**)

- `js/cultivation.js`: `HC.traitMods` wrapper (lines 583–588), `ASTRAL_SUBS` & `BLOOD_SUBS` (lines 317–318), `ELEM.syncSpecialAccess` (lines 320–365), `CULT.pathScores` (lines 609–676)

3.2 Root-Cause Code & Logic Defects

Defect 3A — CRITICAL MATH INVERSION: `op: "MAX_CAP"` on `TR_IN_CRIPPLED_ROOT` Multiplies Cultivation Aptitude by ×62!

In `data/data_core.js`, the curse trait `TR_IN_CRIPPLED_ROOT` (**"ชีพจรรากวิญญาณตีบตัน พิการ — เส้นชีพจรหลักตีบตันแต่กำเนิดทำให้ดูดซับปราณฟ้าดินได้เชื่องช้า แต่กลับมีความอดทนสูงและเหมาะกับการเปลี่ยนไปฝึกวิถีกายา"**) defines:

```
JSON
"modifiers": [
  { "stat": "aptitudes.cultivation", "op": "ADD", "value": -22, "stacking": "SUM"
},
  { "stat": "aptitudes.cultivation", "op": "MAX_CAP", "value": 62, "stacking":
"LOWEST" },
  ...
]
```

Now inspect how **locked** `js/human_core.js:109–115` (`collectMods`) processes every modifier where `m.op !== 'ADD'`:

```
JavaScript
const push = (m, src) => {
  const p = normPath(m.stat);
  if (m.op === 'ADD') adds[p] = (adds[p] || 0) + m.value;
  else (mults[p] = mults[p] || []).push({ v: m.value, grp: m.group || m.stackGroup
|| ('_' + (src || 'x')) });
};
```

And in `HC.traitMods` (`human_core.js:104`): `m.op === 'ADD' ? ... : { stat: m.stat, op: 'MULT', value: 1 + (m.value - 1) * k, group: m.stackGroup }`

- **Catastrophic Result**:
  - Because `m.op` is `'MAX_CAP'` (`!== 'ADD'`), `human_core.js` converts it into **`op: 'MULT', value: 62`**!

- For any NPC with `raw aptitudes.cultivation > 22` (nearly 95% of NPCs!), `(raw - 22) * 62` explodes to `500..4,800`, clamping `HC.eff(npc, 'aptitudes.cultivation')` to **`100`** **(เหนือมนุษย์, 99.9th percentile)**!
- **Chain Reaction**:
    1. Instead of crippling their Qi-gathering aptitude and pushing them into `BODY` (`preferBodyPath: 40`), `TR_IN_CRIPPLED_ROOT` turns them into a **max-100 Qi Cultivation prodigy**!
    2. In `CULT.pathScores`, `XIAN` (`+0.50 * aptitudes.cultivation`) gains `+25` points over `BODY` (which has `0` weight on `aptitudes.cultivation`), so crippled-root mortals systematically **reject `BODY` and pick `XIAN`**!
    3. `HC.derived(npc).cultivationTalent` spikes artificially, causing `BN.daily` (`meritocracy` pill/treasure allocation) to treat a crippled-meridian disciple as the #1 cultivation genius in the sect!

Defect 3B — Hallucinated Constitution IDs in `ELEM.syncSpecialAccess` (`cultivation.js:337–342`)

In `js/cultivation.js`:

```javascript
if (npc.h.constitution === 'CON_STAR_MARROW' || npc.h.constitution === 'CON_VOID_SPIRIT') {
  astralSources.add('CONSTITUTION:' + npc.h.constitution);
}
if (npc.h.constitution === 'CON_BLOOD_FIEND' || npc.h.constitution === 'CON_ASURA_WAR') {
  bloodSources.add('CONSTITUTION:' + npc.h.constitution);
}
```

In `data/data_core.js`, **none of those four IDs exist**. All constitution IDs use the prefix `CONST_`:

- `CONST_STAR_MIND` (กายาจิตดาราหยั่งรู้)
- `CONST_BEAST_BLOOD` (กายาโลหิตอสูรบรรพกาล)
- `CONST_DRAGON_MARROW` (กายาไขกระดูกมังกรสวรรค์)
- Plus bloodline/innate traits in `npc.h.traits`: `TR_BL_BLOOD_CURSE`, `TR_BL_DIVINE_PUPIL`, `TR_IN_ARRAY_GENIUS`, and `TR_GEN_*_6` (Beast Blood) / `TR_GEN_*_7` (Heaven Star) / `TR_GEN_*_18` (Void Emptiness).
- **Result**: No mortal or cultivator in the entire game **ever** unlocks `ASTRAL` or `BLOOD_DEMON` affinity via their innate Constitution! Consequently, when `CULT.pathScores` evaluates `ASTRAL` paths (`ASTRO_DIVINATION`, `VOID_SPACE`, `CHRONO_EPOCH`) or `BLOOD_DEMON` paths (`DEMON`, `DEVOURING_FLESH`,

PRIMORDIAL_SHIFTER) for a mortal (!npc.cult), specialAccess.unlocked is always false (accessFactor = 0.58), artificially penalizing all ASTRAL and BLOOD_DEMON paths during initial path selection!

Defect 3C — Stale / Misspelled Subpath IDs in ASTRAL_SUBS & BLOOD_SUBS (cultivation.js:317–318)

- ASTRAL_SUBS lists 'STAR_CHART_ORACLE', whereas the actual canonical subpath ID in ASTRO_DIVINATION is **'STAR_CHART_PROPHET'** (verified in visitors_clues.js:334 and v240_user_audit_regression.js:183). Thus, STAR_CHART_PROPHET fails ASTRAL_SUBS.has(c.sub)—losing both initial specialMastery.ASTRAL (cultivation.js:392) and SUBPATH:STAR_CHART_PROPHET special access!
- Furthermore, Wave-4 (v2.2.0) subpaths with explicit ASTRAL or BLOOD_DEMON identity (e.g., VOID_HERETIC_PACT, DIVINE_SUMMONER, DEVOURING_FLESH, NETHER_GRAVE, OCCULT_MAGE subpaths) were never audited into ASTRAL_SUBS / BLOOD_SUBS or path-level checks (`if (c.path === 'VOID_HERETIC_PACT' || c.path === 'ASTRO_DIVINATION' || c.path === 'VOID_SPACE' || c.path === 'CHRONO_EPOCH')` is missing—only `c.path === 'ARRAY_TALISMAN'` and `c.path === 'DEMON'` are checked at lines 329–331!).

Defect 3D — Dead behaviorMods & Unconditional condition Trait Modifiers in data_core.js

1. data_core.js defines behaviorMods.decisionBias on curated traits (preferBodyPath: 40, studySword: 35, tacticalArray: 35, inventRecipe: 35, seekSecretRealm: 35, avengeGrudge: 45, defectUnderPressure: 35, forgiveAndMediate: 40), **none of which are read anywhere** in CULT.pathScores, COG, or AI.
2. data_core.js defines condition objects on trait modifiers (maxHealth: 50, maxHealth: 65, requiredSeason: "WINTER", inCombatAgainstGrudge: true), which human_core.js ignores—granting low-HP berserk multipliers (×1.18) at 100% HP and winter debuffs in mid-summer, and applying non-existent stat physiology.poisonResistance (TR_BL_POISON_IMMUNE) instead of physiology.immunity.

3.3 Architectural Remediation Blueprint for AI Developers (Zero human_core.js Edits!)

1. **Intercept Unsupported op & Conditional Modifiers in HC.traitMods Wrapper (cultivation.js:583)**:
    - Without touching human_core.js, enhance the existing HC.traitMods wrapper in cultivation.js:

    - Filter out or translate `m.op === 'MAX_CAP'` (never let `origTraitMods` turn `MAX_CAP: 62` into `MULT: 62`! Since `-22 ADD` already caps a `99` innate roll at `77` and average `50` at `28`, dropping `MAX_CAP` or converting it into a bounded negative `ADD` immediately cures the `×62` bug while keeping `HC.FINGERPRINT` untouched!).
    - Remap `physiology.poisonResistance` -> `physiology.immunity` in `HC.traitMods` (or `data_core.js`).
    - Strip combat/seasonal `m.condition` entries from static `HC.traitMods` (or gate `requiredSeason: 'WINTER'` dynamically on `W.season === 3` + invalidate `npc._c` on season change in `SIM.dailyUpdate`, and evaluate `maxHealth` / `inCombatAgainstGrudge` inside `STATS` / `COMBAT`!).
2. **Fix `ELEM.syncSpecialAccess`, `ASTRAL_SUBS`, and `BLOOD_SUBS`:**
    - Replace dead `CON_*` checks with real `CONST_STAR_MIND`, `CONST_BEAST_BLOOD`, `CONST_DRAGON_MARROW`, and trait checks (`TR_BL_DIVINE_PUPIL`, `TR_BL_BLOOD_CURSE`, `TR_IN_ARRAY_GENIUS`).
    - Add Major Path checks for all Astral/Void/Time paths (`ASTRO_DIVINATION`, `VOID_SPACE`, `CHRONO_EPOCH`, `VOID_HERETIC_PACT`, `PRIMORDIAL_CHAOS`) and Blood/Flesh paths (`DEVOURING_FLESH`, `VENOM_GU`, `PRIMORDIAL_SHIFTER`).
    - Fix `'STAR_CHART_ORACLE'` -> `'STAR_CHART_PROPHET'` and dynamically derive `ASTRAL_SUBS` / `BLOOD_SUBS` at boot by scanning `CULT.subList` for any subpath whose `elementProfile.weights.ASTRAL > 0` or `weights.BLOOD_DEMON > 0`! (This permanently prevents future subpath typos when adding Path #44+!)
3. **Wire `behaviorMods.decisionBias` into `CULT.pathScores` & `COG`:**
    - Read `(npc.h.traits || [])` in `CULT.pathScores`: e.g., `preferBodyPath` boosts `BODY` / `PRIMORDIAL_SHIFTER` / `ARENA_GLORY`; `studySword` boosts Sword subpaths; `tacticalArray` boosts `ARRAY_TALISMAN` / `ASTRO_DIVINATION`.

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4. `[PATH-ALIAS-04]` Subpath Alias Collisions & Silent Identity Erasure

4.1 Exact Code Anchors

- `js/cultivation.js`: `CULT.SUB_ALIASES` & `CULT.resolveSubId` (lines 685–722)
- `js/martial.js`: `profiles` (lines 35–95), `M.normalize` (lines 185–195), `M.SUB_EQUIP_BONUSES` (lines 396–412)

- `js/visitors_clues.js`: `LEGENDARY_WEAPONS`, `LEGENDARY_ARMORS`, `LEGENDARY_ACCESSORIES` `preferredSubs` arrays (lines 75–510)

4.2 Root-Cause Code & Logic Defects

`M.normalize(n)` (`martial.js:187`) mutates `n.cult.sub` in place using `CULT.resolveSubId(c.sub)`:

```javascript
const rSub = CULT.resolveSubId(c.sub);
if (rSub !== c.sub) { c.sub = rSub; if (c.track && c.track.subPath)
c.track.subPath = rSub; }
```

However, `CULT.SUB_ALIASES` (`cultivation.js:685–718`) aliases **dozens of subpath IDs** that are **still actively referenced under their pre-alias names** across `martial.js` and `visitors_clues.js`!

Pre-Alias ID Referenced in Code/Items	**Rewritten By `M.normalize` To (`c.sub`)**	**Broken Downstream System(s)**
`SOUL_MUSICIAN`	`SEVEN_EMOTIONS_ZITHER`	`M.SUB_EQUIP_BONUSES.SOUL_MUSICIAN` (dead code); `LEG_WPN_JIUXIAO_PHOENIX_ZITHER.preferredSubs`
`GRAND_FORMATION_ARCHITECT`	`EARTH_VEIN_WARDEN`	`M.SUB_EQUIP_BONUSES.GRAND_FORMATION_ARCHITECT` (dead code); `LEG_WPN_CHUNQIU_JUDGMENT_BRUSH`, `LEG_ARM_CELESTIAL_CRANE_ROBE`, `LEG_ACC_TAICHI_HEAVEN_SEAL`
`SPIRIT_THREAD_PUPPET`	`THOUSAND_THREAD_ASSASSIN`	`M.SUB_EQUIP_BONUSES.SPIRIT_THREAD_PUPPET` (dead code); `LEG_WPN_TIANCAN_FATE_WHIP`, `LEG_ARM_SHADOW_KIRIN_CUIRASS`

Pre-Alias ID Referenced in Code/Items	Rewritten By `M.normalize` To (`c.sub`)	Broken Downstream System(s)
`GRAND_STRATEGIST`	`BATTLEFIELD_STRATEGIST`	`LEG_WPN_CHUNQIU_JUDGMENT_BRUSH`, `LEG_ACC_TAICHI_HEAVEN_SEAL`
`IMPERIAL_PEACE_SOVEREIGN`	`SECT_STEWARD`	`LEG_ACC_TAICHI_HEAVEN_SEAL`
`IRON_COLOSSUS` / `FORTRESS_BASTION_ARCHITECT`	`HEAVENLY_GOLEM`	`M.PROFILES` distinct profiles erased; `combat.js:1214` check; `LEG_ARM_IMMORTAL_TORTOISE_PLATE`
`SYMBIOTIC_FANG_FUSION`	`SYMBIOTIC_BEAST_WARRIOR`	`M.PROFILES` distinct entry; `LEG_ARM_SHADOW_KIRIN_CUIRASS`, `LEG_ACC_DRAGON_PHOENIX_BANGLE`
`PILL_FLAME_SOVEREIGN`	`PILL_FURNACE_IMMORTAL`	`LEG_ACC_DRAGON_PHOENIX_BANGLE`
`RED_THREAD_ARBITER` / `THOUSAND_THREAD_PUPPETEER`	`FATE_SEVERING_THREAD` / `THOUSAND_THREAD_ASSASSIN`	`M.PROFILES`; `LEG_WPN_TIANCAN_FATE_WHIP`, `LEG_ACC_SAMSARA_KARMA_RING`
`THUNDER_CLOUD_CALLER` (in `LEG_WPN_BAJIAO_TEMPEST_FAN`)	Actual subpath in `CELESTIAL_WEATHER`: `THUNDERSTORM_CALLER`	`LEG_WPN_BAJIAO_TEMPEST_FAN.preferredSubs` never matches!
`GOLDEN_LOTUS_CHAN` (in `LEG_WPN_BODHI_NIRVANA_ROSARY`)	Actual subpath in `BUDDHA_DHARMA`: `GOLDEN_LOTUS_ARHAT`	`LEG_WPN_BODHI_NIRVANA_ROSARY.preferredSubs` mismatch!

- **Chain Reaction**:
  - In `visitors_clues.js:559`, matching `def.preferredSubs.includes(n.cult.sub)` awards **+135 gear**

suitability score (deciding who in the sect receives or buys a unique Legendary Relic!). Because `preferredSubs` contains un-normalized or misspelled subpath IDs, disciples on the actual canonical subpaths (`EARTH_VEIN_WARDEN`, `BATTLEFIELD_STRATEGIST`, `SECT_STEWARD`, `HEAVENLY_GOLEM`, `THUNDERSTORM_CALLER`, `GOLDEN_LOTUS_ARHAT`, etc.) lose `+135` suitability on their own signature Legendary Relics, causing relics to be misallocated or hoarded by off-spec elders!
- Similarly, `M.SUB_EQUIP_BONUSES` silently drops custom weapon/armor resonance bonuses for aliased subpaths, falling back to generic `+6%` defaults.

4.3 Architectural Remediation Blueprint for AI Developers

1. **Canonicalize All Subpath Lookups at Registration Time**:
   - At module initialization in `martial.js` and `visitors_clues.js`, normalize all keys of `M.SUB_EQUIP_BONUSES`, `M.PROFILES`, `M.PASSIVE_SLOT_BONUSES`, and every `preferredSubs` / `subpaths` array on items, resonance sets, and techniques through `CULT.resolveSubId(subId)`!
   - Fix all hallucinated subpath names in `LEGENDARY_EQUIPMENT` (`THUNDER_CLOUD_CALLER` -> `THUNDERSTORM_CALLER`, `GOLDEN_LOTUS_CHAN` -> `GOLDEN_LOTUS_ARHAT`, etc.) and add a schema validation assertion in `tools/validate_data_schema.js` requiring every `preferredSubs` / `reqSubpath` / `compatibleSubs` entry across all items and techniques to exist in `CULT.subList`!

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5. `[PATH-RET-05]` Broken Retraining Track Initialization, Intra-Path Lifespan Death-Trap, Residual State Pollution & Flat Subpath Tiering

5.1 Exact Code Anchors

- `js/retraining.js`: `M.retrainingStage` (line 5), `M.retrain` (lines 24–41), `M.evaluateRetraining` (lines 44–62)
- `js/knowledge.js`: `branches` (line 116), `M.rollHierarchicalDiscovery` (lines 242–249)
- `js/fishing.js`: `SUBPATH_LOOT_TIER` (lines 406–420), `FISHING.openContainer` (lines 595–610)

5.2 Root-Cause Code & Logic Defects

Defect 5A — `M.retrain` Fails to Initialize `c.track` for Non-`EXPANSION_TRACK_IDS` Subpaths (`retraining.js:30`)

Look at `js/retraining.js:30`:

```
JavaScript
if (CULT.isExpansionTrack ? CULT.isExpansionTrack(k.id) :
(CULT.EXPANSION_TRACK_IDS && CULT.EXPANSION_TRACK_IDS.has(k.id)))
  n.cult.track = CULT.initTrack(k.id, 1, W.day);
```

Compare this with `sim.js:107` and `knowledge.js:150`, both of which use **`if (CULT.trackDef(k.id)) n.cult.track = CULT.initTrack(k.id, 1, W.day);`**.

- **Impact**: Since `v2.0.0`, **all 286 subpaths** have 9-realm tracks in `CULT.trackDef(subId)`, whereas `isExpansionTrack` / `EXPANSION_TRACK_IDS` was a legacy check for the 69 v1.11 expansion tracks. Whenever an NPC retrains into a subpath not covered by `isExpansionTrack`, **`n.cult.track` is left `undefined`**—permanently locking that NPC out of specialization track realms, track realm buffs, and `subpathTraits`!

Defect 5B — The Intra-Path Lifespan Death-Trap (`M.retrainingStage` in `retraining.js:5`)

```
JavaScript
M.retrainingStage = (old, pathChanged) => pathChanged ? 1 : Math.max(1, old -
Math.max(2, Math.ceil(old / 2)));
```

- **Why This Makes Zero Sense in Xianxia Context**:
  - Suppose a 420-year-old Stage 8 `XIAN` Mahayana Elder (`lifespanMax = 1,050` years) currently on `SWORD_IMMORTAL` acquires a manual for `PURE_SWORD_SAINT` or `MIND_SWORD_TELEKINESIS` (**same Major Path `XIAN`**—their Dantian, Qi nature, and Nascent Soul/Mahayana foundation are still `XIAN`!).
  - `M.retrainingStage(8, false)` returns **`8 - 4 = Stage 4 (Golden Core)`**, whose `lifespanMax` is only **`240` years**!
  - Because `ageYears (420) > 240 - 8`, `ageRisk = 100` (`retraining.js:55`), **permanently forbidding every senior cultivator above ~230 years old from ever refining or pivoting their subpath even within their own Major Path**! And if triggered via `M.retrain` directly, they

instantly exceed 100% of their new `lifespanMax` (`420 > 240`) and **drop dead of "old age" (ชราภาพ) in `SIM.dailyUpdate` within days**!

- **Contextual Distinction Needed**:
  - **Cross-Path Conversion (`pathChanged === true`, e.g., `XIAN` -> `BODY` or `SHADOW_SAMURAI`)**: Truly dispersing one's Dantian to rebuild a completely different cultivation system (`newStage = 1` or `2`, retaining `lifeBonus` or partial physical/soul longevity cushion so mid-aged elders don't evaporate instantaneously unless reckless).
  - **Intra-Path Subpath Pivot (`pathChanged === false`, e.g., `XIAN/SWORD_IMMORTAL` -> `XIAN/PURE_SWORD_SAINT`)**: Should preserve Major Realm Dantian (`newStage = Math.max(1, old - 1)` or `old` with `prog = 0` + `retrainUntil` foundation-stabilization debuff) while resetting `c.track` (`ordinal = 1`) and deactivating old exclusive subpath passives/innates!

Defect 5C — Residual Path-Specific State Pollution After `M.retrain` (`retraining.js:27–38`)

When `M.retrain` replaces `n.cult`, it leaves behind stale top-level properties on `n` from the previous Major Path:

1. **`n.pathGauge`**: An NPC retraining from `SHADOW_SAMURAI` (`IAI_FOCUS = 100`) into `XIAN` keeps `n.pathGauge.id === 'IAI_FOCUS'` until overwritten, or keeps `WANDERING_DAO` `ROUTE_INSIGHT` / `VENOM_GU` `SYMBIOTIC_GU` gauges across paths!
2. **`n.mechState` & `n.karmaState`**: If leaving `MECH_PUPPET` or `KARMA_SAMSARA`, old construct/karma objects persist on `n`.
3. **`n.secondaryResources` & `n.demonicRealmsGauge`**: Secondary gauges from the old path remain populated (`resolveMultiGaugeState` in `stats.js:383` continues granting `secondaryBuffs`!).
4. **Era-5 States (`n.qinggongState`, `n.mantraState`, `n.defensiveArtState`, `n.supportState`, `n.forbiddenArtState`)**: Retain `preferredMode` / `preferredArchetype` from the old path (e.g., someone retraining **into** `SHADOW_SAMURAI` keeps `preferredMode: 'CELESTIAL_LEVITATION'`!).
5. **Equipped Incompatible Gear**: `M.retrain` does **not** unequip items in `n.equip` whose `compatiblePath` / `reqPath` no longer matches `k.path` (or non-Sword/Blade weapons when retraining into `SHADOW_SAMURAI`), leaving unusable dead items stuck in `n.equip` until a shop/bag swap occurs!

Defect 5D — Flat `Tier 1–2` Subpath Pricing (`knowledge.js:116`) & 37-of-286 Sunken Chest Loot Table (`fishing.js:406`)

1. In `knowledge.js:116`: `tier: JD.MARTIAL.subPaths.some(x => x.id === s.id) ? 2 : 1` Every single one of the 286 subpaths is hardcoded to Tier `1` or `2`. Cosmic/Apex subpaths (`CREATION_DESTRUCTION_SAGE`, `ETERNAL_KALPA_SAGE`,

`HUN_DUN_ORIGIN_DEVOURER`) have the exact same `tier`, `studyFee`, `crystalPrice`, and `SECT.canLearnTier` gate (Rank 1 outer disciple!) as basic starter subpaths!

2. In `fishing.js:406–420`, `SUBPATH_LOOT_TIER` only maps 37 Wave-1 subpaths; **all 249 Wave-2/3/4 subpaths fall back to `|| 1` (`Common`)**, meaning 87% of all Grand Dao subpaths flood the lowest-rarity Tier-1 loot pool (`56%` weight), while Tier 2/3/4 chests can only ever roll the 22 old v1.x subpaths!

5.3 Architectural Remediation Blueprint for AI Developers

1. **Fix `M.retrain` State Cleanup & Track Initialization**:
   - Change `retraining.js:30` to `if (CULT.trackDef(k.id)) n.cult.track = CULT.initTrack(k.id, 1, W.day);`.
   - On `pathChanged === true`: reset/re-initialize `n.pathGauge` (`CULT.ensurePathGauge(n)`), clear `n.mechState` (if `k.path !== 'MECH_PUPPET'`), `n.karmaState` (if `k.path !== 'KARMA_SAMSARA'`), reset incompatible `n.secondaryResources`, reset Era-5 `preferredMode`/`preferredArchetype` to `'AUTO'`, and unequip any `n.equip[slot]` failing `M.gearUsable(n, it)` into `n.inv.gear`.
2. **Reform `M.retrainingStage(old, pathChanged)`**:
   - Intra-path subpath switch (`pathChanged === false`): `Math.max(1, old - 1)` (at most 1 minor realm regression + reset `c.track` to ordinal 1 + 20-day meridian stabilization cooldown), so mid/high-realm elders can realistically pivot subpaths within their own Dao without dying of old age.
   - Cross-path switch (`pathChanged === true`): retain residual physical/soul longevity buffer (`lifeBonus` floor from prior apex body/soul refinement) or evaluate `ageRisk` strictly before execution.
3. **Derive Subpath Tier Dynamically Across All 286 Subpaths**:
   - Compute subpath `tier` (`1..5`) and `lootTier` (`1..4`) deterministically from the subpath's authored `combatBonus + defenseBonus + breakthroughMod`, special-element profile (`ASTRAL`/`BLOOD_DEMON`/composite), or explicit `sub.tier` metadata across all 43 paths—eliminating the 37-subpath hardcode in `fishing.js` and flat Tier 1–2 pricing in `knowledge.js`.

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6. Mandatory End-to-End Verification Suite Spec (For Part 1 Fixes)

When GPT or Gemini implements Part 1 remediations, it must add `test/v270_paths_subpaths_part1_autonomy_suite.js` testing **real multi-tick simulation behavior**:

1. **`test_crippled_root_no_mult62_explosion`**: Generate an NPC with `TR_IN_CRIPPLED_ROOT` (`raw aptitudes.cultivation = 55`). Assert `HC.eff(npc, 'aptitudes.cultivation') <= 40` (never `100`!), `HC.FINGERPRINT === '155:2f9c9dd7'`, and `CULT.majorPathScores(npc)[0].path` favors `BODY` / physical paths over `XIAN`.
2. **`test_special_element_constitution_and_all_286_subpaths`**: Verify `CONST_STAR_MIND` unlocks `specialAccess.ASTRAL`, `CONST_BEAST_BLOOD` / `TR_BL_BLOOD_CURSE` unlocks `BLOOD_DEMON`, and `STAR_CHART_PROPHET` receives `ASTRAL` unlock & starting mastery.
3. **`test_zero_dead_aliases_across_items_and_bonuses`**: Iterate all `LEGENDARY_EQUIPMENT` `preferredSubs` and `M.SUB_EQUIP_BONUSES` keys; assert 100% match canonical `c.sub` values after `M.normalize(npc)`.
4. **`test_autonomous_elder_codifies_new_path_primer`**: Place a loyal Stage 6 `CHRONO_EPOCH` elder in a sect whose `unlockedPaths` is `['XIAN']`. Advance `SIM.tick` autonomously (zero UI/editor calls); verify the elder transcribes `CHRONO_EPOCH` into `W.sect.pathLibrary.unlockedPaths` with valid stone/material debit, enabling a newly recruited mortal to study and awaken into `CHRONO_EPOCH`.
5. **`test_unified_path_library_and_retrain_clean_state`**: Verify `M.retrain` initializes `c.track` for all 286 subpaths, cleans `n.pathGauge`/`mechState`/`secondaryResources`/incompatible `equip`, and allows a 400-year-old Stage 8 `XIAN` elder to pivot from `SWORD_IMMORTAL` to `PURE_SWORD_SAINT` without dying of instant lifespan collapse.